Title Page (Unblinded for Editorial Office Only)

The Epistemic Fluency Trap in AI-Augmented Security Operations: Generative Rationale Fluency, Contested Evidence Cues, and Analyst Verification

Tran Duc Le
Corresponding Author

Submitted to *Journal of Management Information Systems* (Taylor & Francis)

October 2026

Running Head: The Epistemic Fluency Trap in AI-Augmented SOCs

Target Journal: *Journal of Management Information Systems* (Taylor & Francis; Print ISSN: 0742-1222, Online ISSN: 1557-928X)

CRediT Author Contributions Statement

Tran Duc Le: Conceptualization, Methodology, Software, Validation, Formal Analysis, Investigation, Data Curation, Writing - Original Draft, Writing - Review & Editing, Visualization, Project Administration.

Data and Code Availability Statement

All source code, the `SOC-ITPI-24` stimulus benchmark, multi-corpus empirical evaluation pipelines, empirical log-likelihood tables, and trace-driven decision-theoretic sensitivity scripts supporting the findings of this study are openly available in an anonymous repository for double-blind peer review and a public repository for camera-ready archival:

- **Anonymous Double-Blind Review Repository:** https://anonymous.4open.science/r/epistemic-fluency-trap/
- **Camera-Ready Public Repository:** https://github.com/tranducle/epistemic-fluency-trap

Funding Statement

This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors.

Conflict of Interest Disclosure

The author declares no conflicts of interest.

Ethics Statement

This study did not involve any human participants, human-subject experiments, surveys, interviews, or animal subjects. All empirical evaluations and trace-driven decision-theoretic analyses were conducted exclusively on publicly available, de-identified cybersecurity telemetry datasets (`OTRF Security-Datasets`, `SALAD-SOC`, `LMD-2023`, and `CISA KEV`). Accordingly, institutional review board (IRB) human-subjects approval and informed consent were not applicable.

Acknowledgments

The author thanks the maintainers and open-source contributors of the Open Threat Research Forge (`OTRF Security-Datasets`), `SALAD-SOC`, `LMD-2023`, and the Cybersecurity and Infrastructure Security Agency Known Exploited Vulnerabilities (`CISA KEV`) public cybersecurity repositories for making their telemetry corpora openly available.

Declaration of Generative AI and AI-Assisted Technologies

During the preparation of this work, the author used generative AI and AI-assisted tools strictly for grammar checking, academic tone polishing, LaTeX formatting, drawing LaTeX/TikZ figures, and code review. After using these tools, the author reviewed, verified, and edited all content and code as needed and takes full responsibility for the integrity, accuracy, and scientific claims of the publication.